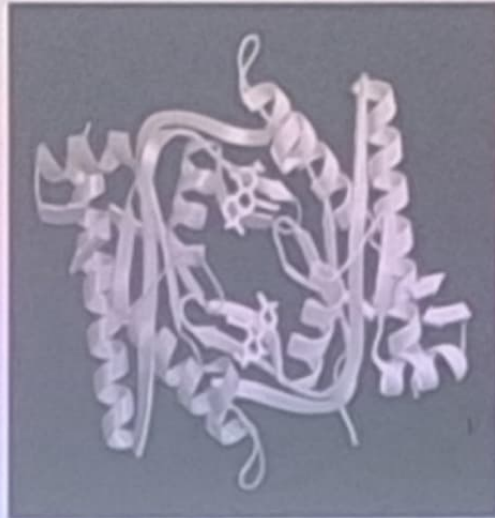


Transport Proteins

- Bind and carry molecules or ions to organs in the blood plasma.
- **Lipoproteins** in blood plasma carries lipids from one organ to other organs.

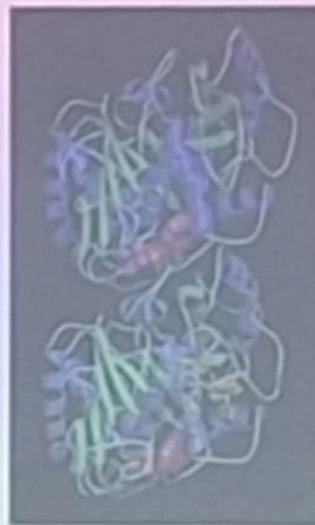


Nutrient and Storage Proteins

- Seeds of many plants store nutrient proteins required for the growth of the germinating seedlings.
- Ovalbumin, the major protein of egg white, and casein the major protein of milk are examples of nutrient proteins.

Contractile or Motile Proteins

- Some proteins endow cells and organisms with the ability to contract, to change shape, or to move.
- Tubulin is the protein from which microtubules are built.



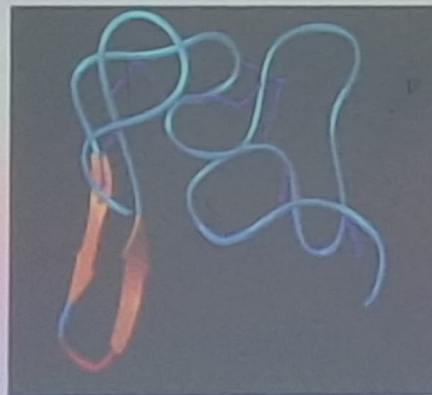
Structural Proteins

- Many proteins serve as supporting filaments, cables, or sheets, to give biological structures strength or protection.
- Major component of tendons and cartilage is the fibrous protein collagen, which has very high tensile strength.
- Ligaments contain elastin, a structural protein.



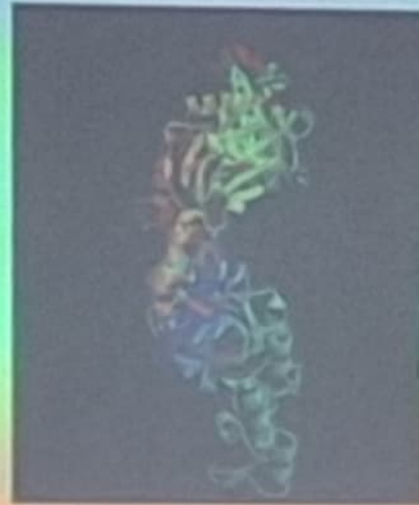
Defense Proteins

- Defend organisms against invasion by other species or protect them
- Immunoglobulin or antibodies, are made by the lymphocytes of vertebrates and can recognize & precipitate or neutralize invading bacteria
- Fibrinogen and thrombin are blood clotting proteins



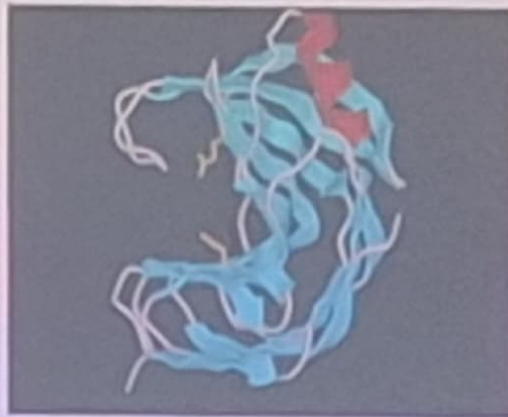
Regulatory Proteins

- Help regulate cellular or physiological activity.
- The cellular response to many hormonal signals is often mediated by a class of GTP-binding proteins called G proteins.



Enzymes

- Most varied and specialized **proteins** with catalytic activity.
- All chemical reactions of organic biomolecules in cells are catalyzed by enzymes.



Chemical Reactions

- Chemical reaction – is a process that changes one set of chemicals into another set of chemicals.
- The elements or compounds that enter into a chemical reaction are the reactants.
- The elements or compounds produced by a chemical reaction are known as products.

Energy in Reactions

- Energy is neither created nor destroyed, therefore energy is released or absorbed whenever chemical bonds form or are broken.
- Chemical reactions that release energy often occur spontaneously.
Chemical reactions that absorb energy will not without a source of energy.

Activation Energy

Do all reactions occur spontaneously??

- Activation energy is the minimum amount of energy that must be available to reactants for a chemical reaction to occur. The activation energy (E_a) of a reaction is measured in kilojoules per mole (kJ/mol) or kilocalories per mole (kcal/mol).

Some chemical reactions that make life possible are too slow or have activation energies that are too high to make them practical for life.

These reactions require catalysts or substances that speed up the rate of a chemical reaction.

CATALYSTS

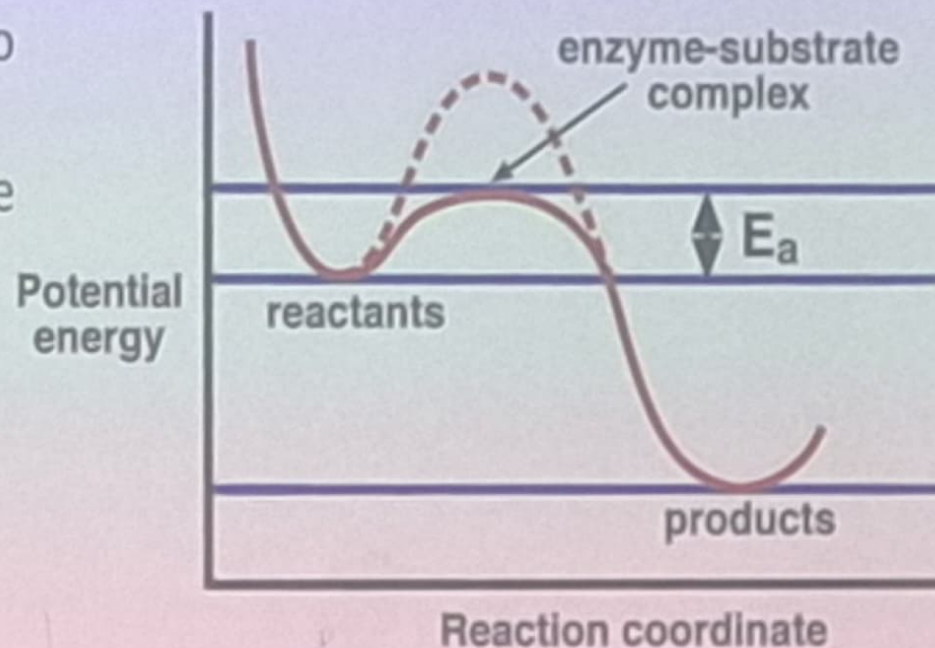
Catalysts work by lowering a reaction's activation energy.

ENZYMES

A biomolecule either Protein or RNA, that catalyzes a specific chemical reaction, enhances the rate of a reaction by providing a reaction path with a lower activation energy.

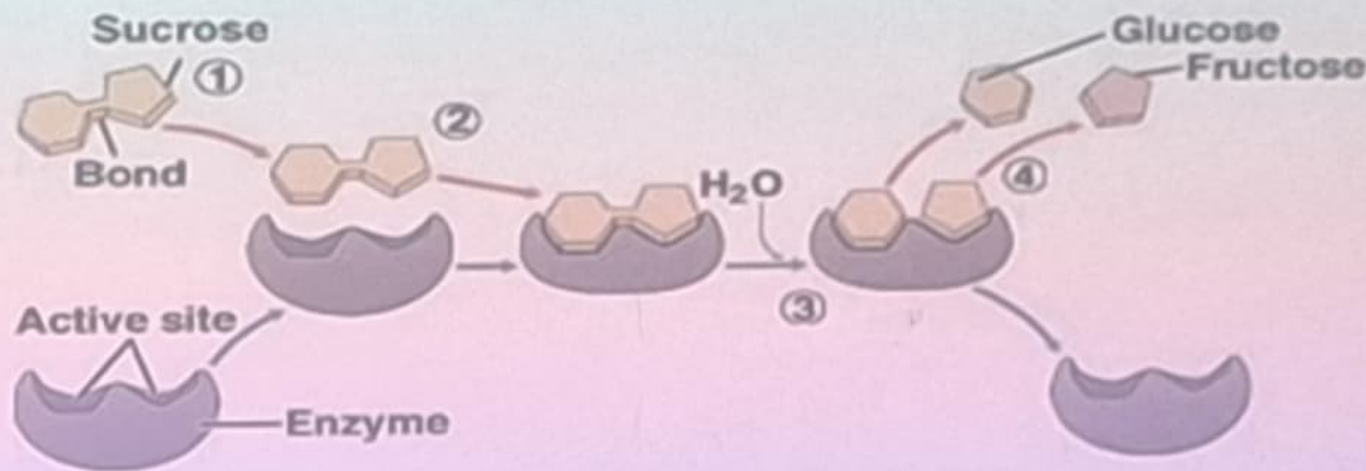
In order to stay alive, organisms need to carry out reactions that require energy.

- For a chemical reaction to occur, the reactants must collide with enough energy so that existing bonds will be broken and new bonds will be formed.
- Enzymes provide a place where reactants can be brought together to react.
- The reactants of enzyme – catalyzed reactions are known as substrates



The Enzyme/Substrate Complex

- Each protein (enzyme) has a specific shape, the substrate binds to the active site on the protein to help speed up the reaction.
- Also known as..... “lock and key”



Fundamental Properties

- 1) Catalytic power-speed up reactions 10^8 to 10^{20} fold.
They speed up reactions without being used up.

Catalytic Power

Catalytic Power is defined as the Ratio of the Enzyme-Catalyzed Rate of a Reaction to the Uncatalyzed Rate

2) Specificity

- a) for *substrate* - ranges from absolute to relative
- b) for *reaction catalyzed*

Specificity

Defined as the Selectivity of Enzymes for the Reactants Upon which They Act

3) Regulated-- some enzymes can sense metabolic signals

Regulation

Regulation of enzyme activity ensures that the rate of metabolic reactions is appropriate to cellular requirements

- essential to the integration and regulation of metabolism

Achieved by various ways

- Inhibitor
- Activator
- Hormonal
- Rate of synthesis

In an enzyme-catalyzed reaction, none of the substrate is diverted into nonproductive side of reactions, so no wasteful by-products are produced.

The substances upon which an enzyme acts are traditionally called- **substrates**

The selective qualities of an enzyme are collectively recognized- **specificity**

The specific site on the enzyme where substrate binds and catalysis occurs is called- **active site**

Enzymes are Proteins that act as Biological Catalyst

They speed up rate of chemical reactions w/o being consumed in process.

